

CUSTOMER BEHAVIOUR ANALYSIS

PROJECT REPORT

**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING**

PREPARED BY

SUNILKUMAR M

Data Preparation • SQL Analysis • Power BI Dashboard

CERTIFICATE

This is to certify that the project titled “Customer Behaviour Analysis” has been prepared as a data analytics project using Python, Pandas, PostgreSQL, SQL, Power BI and DAX. The project demonstrates an end-to-end workflow for preparing customer shopping data, extracting business insights and presenting those insights through an interactive dashboard.

Project Details	Information
Project Title	Customer Behaviour Analysis
Project Type	Data Analytics
Primary Dataset	customer_shopping_behavior.csv
Database	PostgreSQL
Database Tool	pgAdmin 4
Visualization	Microsoft Power BI
Prepared By	Sunil Kumar M

DECLARATION

I am **Mr. SUNILKUMAR M** ,hereby declare that this project report titled “**Customer Behaviour Analysis**” is prepared as part of my data analytics project work. The report describes the data preparation, transformation, database integration, SQL analysis and Power BI dashboard workflow implemented for the customer shopping behaviour dataset.

DATE :

1.

PLACE : CHENNAI

SIGNATURE OF CANDIDATES

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the completion of this project. I am thankful to my trainers, mentors, friends and family for their guidance and encouragement. I also acknowledge the tools and technologies used in this project, including Python, Pandas, PostgreSQL, SQL and Power BI.

ABSTRACT

Customer Behaviour Analysis is an end-to-end data analytics project designed to understand customer purchasing patterns, loyalty, satisfaction, discounts, product preferences, subscription behaviour and demographic contribution to sales. The project begins with the `customer_shopping_behavior.csv` dataset and follows a structured workflow: data preparation in Python/Pandas, storage in PostgreSQL, business analysis using SQL, and visualization in Power BI using DAX.

During preprocessing, the dataset was inspected, missing review ratings were handled using category-wise median imputation, column names were standardized, and additional analytical features such as `age_group` and `purchase_frequency_days` were created. Redundant information between `discount_applied` and `promo_code_used` was checked and the redundant `promo_code_used` field was removed.

The cleaned dataset was loaded into a PostgreSQL database named `customer_behavior` using SQLAlchemy and psycopg2. Ten SQL business questions were executed to analyse revenue by gender, discounted high-value purchases, product ratings, shipping types, subscription spending, discount rates, customer segments, top products by category, repeat-buyer subscription behaviour and revenue contribution by age group.

The final stage connected PostgreSQL to Power BI. DAX measures and interactive visualizations were used to build a customer behaviour dashboard. The project demonstrates how raw customer data can be converted into structured, business-oriented insights that support marketing, product and customer-engagement decisions.

INTRODUCTION

In today's competitive retail environment, understanding customer behaviour is essential for improving sales, customer satisfaction, and business growth. Customers have different purchasing habits, product preferences, spending patterns, responses to discounts, shipping preferences, and levels of loyalty. Analysing this information helps businesses understand what influences customers when making purchasing decisions.

The Customer Behaviour Analysis project focuses on analysing customer shopping data to identify meaningful patterns and trends in customer purchasing behaviour. The project uses a complete data analytics workflow, starting from raw data collection and preparation and ending with an interactive Power BI dashboard.

Initially, the `customer_shopping_behavior.csv` dataset was collected and imported into Jupyter Notebook using Python and Pandas. The dataset was explored to understand its structure, data types, statistical information, and missing values. Data cleaning was then performed to improve the quality of the dataset. Missing review ratings were handled using category-wise median values, column names were standardized, and unnecessary or duplicate information was removed.

After cleaning, additional features such as Age Group and Purchase Frequency in Days were created to make the data more useful for analysis. The transformed dataset was then connected to a PostgreSQL database created using pgAdmin 4. SQLAlchemy and the PostgreSQL driver were used to load the cleaned dataset into the database.

Once the data was stored in PostgreSQL, SQL queries were executed to answer important business questions related to revenue, discounts, product ratings, shipping methods, subscriptions, customer loyalty, product popularity, repeat purchases, and age-group revenue contribution.

Finally, the PostgreSQL database was connected to Microsoft Power BI. DAX functions and measures were used to calculate important Key Performance Indicators (KPIs), and interactive charts and visualizations were created to present the analysis in an easy-to-understand format.

The main purpose of this project is to convert raw customer shopping data into meaningful business insights that can help the company improve customer engagement, optimize marketing campaigns, understand customer loyalty, improve product strategies, and make data-driven business decisions.

1. BUSINESS PROBLEM STATEMENT

A leading retail company wants to better understand its customers' behaviour in terms of sales, purchasing patterns, customer loyalty, satisfaction and demographics. The management team has noticed changes in customer behaviour across product categories and is interested in understanding how discounts, reviews, purchase frequency, subscriptions and customer characteristics influence purchasing decisions.

The company wants to leverage customer shopping data to identify actionable patterns and improve customer engagement, marketing effectiveness and product strategies.

Overarching business question:

“How can the company leverage consumer shopping data to identify opportunities, improve customer engagement, and optimize marketing and product strategies?”

1.1. Business Context

- Understand which customer groups contribute most to revenue.
- Identify purchasing and product-preference patterns.
- Measure the relationship between discounts and high-value purchases.
- Understand customer loyalty through previous-purchase behaviour.
- Compare subscribers and non-subscribers.
- Identify highly rated and frequently purchased products.
- Understand revenue contribution across age groups.
- Provide an interactive dashboard for management decision-making.

2. BUSINESS QUESTIONS

1. What is the total revenue generated by male versus female customers?
2. Which customers used a discount but still spent more than the average purchase amount?
3. Which are the top five products with the highest average review rating?
4. How do average purchase amounts compare between Standard and Express Shipping?
5. Do subscribed customers spend more than non-subscribers?
6. Which five products have the highest percentage of purchases with discounts applied?
7. How can customers be segmented into New, Returning and Loyal based on previous purchases?
8. What are the top three most purchased products within each category?
9. Are repeat buyers with more than five previous purchases more likely to subscribe?
10. What is the revenue contribution of each age group?

3. PROJECT DELIVERABLES

Deliverable	Description
Data Preparation & Modeling	Clean and transform the raw dataset using Python and Pandas.
Data Analysis (SQL)	Organize structured customer transactions and execute business queries for insights.
Visualization (Power BI)	Build an interactive dashboard showing key metrics, patterns and trends.
Report & Presentation	Document findings and provide actionable business recommendations.
GitHub Repository	Maintain Python scripts, SQL queries, dashboard-related files and project documentation in a structured repository.

4. PROJECT OBJECTIVES

- Prepare raw customer shopping data for analysis.
- Improve data quality through systematic cleaning.
- Create useful derived features for customer analysis.
- Store the cleaned dataset in a relational database.
- Use SQL to answer practical business questions.
- Create DAX measures for dashboard KPIs.
- Visualize customer behaviour in Power BI.
- Generate business recommendations from the analysis.

5. DATASET DESCRIPTION

The primary input file is `customer_shopping_behavior.csv`. It contains customer and transaction-level attributes suitable for analysing purchasing behaviour, product performance, customer loyalty, discounts, shipping, ratings and subscriptions.

Field / Concept	Use in Analysis
customer_id	Identify customers and calculate customer counts.
age	Demographic analysis and age-group creation.
gender	Gender-wise revenue comparison.
category	Category-level analysis and rating imputation.
item_purchased	Product performance and purchase analysis.
purchase_amount	Revenue and spending analysis.
review_rating	Product/customer satisfaction analysis.
shipping_type	Standard vs Express spending comparison.
subscription_status	Subscriber vs non-subscriber analysis.
previous_purchases	Customer loyalty and segmentation.
frequency_of_purchases	Purchase-frequency analysis.
discount_applied	Discount behaviour analysis.
age_group	Derived demographic segment.
purchase_frequency_days	Derived numerical purchase-frequency measure.

6. TECHNOLOGY STACK

Technology	Role in Project
Python	Programming and data-processing language.
Pandas	Data cleaning, transformation and feature engineering.
Jupyter Notebook	Interactive development and analysis environment.
PostgreSQL	Relational database for cleaned data.
pgAdmin 4	PostgreSQL database administration.
SQL	Business analysis and insight extraction.
SQLAlchemy	Database connection from Python.
psycopg2	PostgreSQL driver for Python.
Power BI	Interactive visualization and dashboarding.
DAX	Calculated measures and analytical KPIs.

7. DATA PREPARATION AND CLEANING USING PYTHON

The first technical stage was importing and exploring the raw dataset in Jupyter Notebook. Pandas was used because it provides efficient DataFrame operations for inspection, missing-value treatment, column transformation and feature engineering.

7.1 Import Dataset

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

7.2 Initial Data Inspection

```
df.head()
df.info()
df.describe(include='all')
df.isnull().sum()
```

These commands were used to understand the first records, column data types, descriptive statistics and missing values.

7.3 Missing Review Ratings

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

Missing review ratings were filled using the median review rating of the corresponding category. Category-wise imputation preserves category-specific rating behaviour better than applying one global median.

7.4 Standardizing Column Names

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
```

7.5 Renaming Purchase Amount

```
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

8. FEATURE ENGINEERING

Feature engineering creates new analytical variables from existing fields. Two important derived features were created in this project.

8.1 Age Group

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']  
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

The `qcut` operation divides customers into four quantile-based groups. This allows revenue and customer behaviour to be compared across demographic segments.

8.2 Purchase Frequency in Days

```
frequency_mapping = {  
    'Fortnightly': 14,  
    'Weekly': 7,  
    'Monthly': 30,  
    'Quarterly': 90,  
    'Bi-Weekly': 14,  
    'Annually': 365,  
    'Every 3 Months': 90  
}  
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

This converts categorical frequency values into approximate numerical intervals, making frequency easier to compare and use in later analysis.

8.3 Redundancy Check

```
(df['discount_applied'] == df['promo_code_used']).all()  
df = df.drop('promo_code_used', axis=1)
```

The project checked whether `discount_applied` and `promo_code_used` carried identical information. Because the information was redundant, `promo_code_used` was removed from the analytical dataset.

9. POSTGRESQL DATABASE INTEGRATION

After cleaning and transformation, a PostgreSQL database named `customer_behavior` was created using pgAdmin 4. The cleaned Pandas DataFrame was then connected to PostgreSQL through SQLAlchemy and the PostgreSQL driver.

9.1 Database Connection

```
connection_url = URL.create(
    "postgresql+psycopg2",
    username=username,
    password=password,
    host=host,
    port=port,
    database=database
)
engine = create_engine(connection_url)
```

For security, the database password used during development is not reproduced in this report.

9.2 Load DataFrame into PostgreSQL

```
table_name = "mytable"

df.to_sql(
    table_name,
    engine,
    if_exists="replace",
    index=False
)
```

The cleaned DataFrame was loaded into the `mytable` table. The `if_exists='replace'` option replaces the existing table during development so that the database reflects the latest cleaned DataFrame.

9.3 Database Workflow

Raw CSV → Pandas DataFrame → Cleaning → Feature Engineering → SQLAlchemy Connection → PostgreSQL `customer_behavior` database → `mytable`

10. SQL BUSINESS ANALYSIS

After loading the cleaned data into PostgreSQL, SQL queries were executed to answer the ten defined business questions. PostgreSQL SQL was used; these are not MySQL queries.

10.1 Business Question 1

1. What is the total revenue generated by male versus female customers?

```
SELECT gender, SUM(purchase_amount) AS total_revenue
FROM mytable
GROUP BY gender;
```

Data Output			Messages	Notifications
	gender text	total_revenue numeric		
1	Female	75191		
2	Male	157890		

10.2 Business Question 2

2. Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id, item_purchased, purchase_amount, discount_applied
FROM mytable
WHERE discount_applied = 'Yes'
AND purchase_amount > (SELECT AVG(purchase_amount) FROM mytable)
ORDER BY purchase_amount DESC;
```

Data Output					Messages	Notifications
	customer_id bigint	item_purchased text	purchase_amount bigint	discount_applied text		
1	249	Belt	100	Yes		
2	1457	Coat	100	Yes		
3	770	Socks	100	Yes		
4	616	Sandals	100	Yes		
5	519	Blouse	100	Yes		
6	862	Skirt	100	Yes		
7	205	Sneakers	100	Yes		
8	1480	Coat	100	Yes		
9	96	Sneakers	100	Yes		
Total rows: 839					Query complete 00:00:00.090	CRLF Ln 21, Col 1

10.3 Business Question 3

3. Which are the top five products with the highest average review rating?

```
SELECT item_purchased,  
       ROUND(AVG(review_rating::numeric), 2) AS average_rating  
FROM mytable  
GROUP BY item_purchased  
ORDER BY AVG(review_rating) DESC  
LIMIT 5;
```

Data Output Messages Notifications		
		
	item_purchased text	average_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

10.4 Business Question 4

4. How do average purchase amounts compare between Standard and Express Shipping?

```
SELECT shipping_type,  
       ROUND(AVG(purchase_amount), 2) AS average_purchase_amount  
FROM mytable  
WHERE shipping_type IN ('Standard', 'Express')  
GROUP BY shipping_type  
ORDER BY average_purchase_amount DESC;
```

Data Output Messages Notifications		
		
	shipping_type text	average_purchase_amount numeric
1	Express	60.48
2	Standard	58.46

10.5 Business Question 5

5. Do subscribed customers spend more than non-subscribers?

```
SELECT subscription_status,  
       ROUND(AVG(purchase_amount), 2) AS average_spend,  
       SUM(purchase_amount) AS total_revenue  
FROM mytable  
GROUP BY subscription_status  
ORDER BY average_spend DESC;
```

	subscription_status text	average_spend numeric	total_revenue numeric
1	No	59.87	170436
2	Yes	59.49	62645

10.6 Business Question 6 – Discount Rate by Product

6. Which five products have the highest percentage of purchases with discounts applied?

```
SELECT item_purchased,  
       ROUND(100.0 * SUM(  
         CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END  
       ) / COUNT(*), 2) AS discount_percentage  
FROM mytable  
GROUP BY item_purchased  
ORDER BY discount_percentage DESC  
LIMIT 5;
```

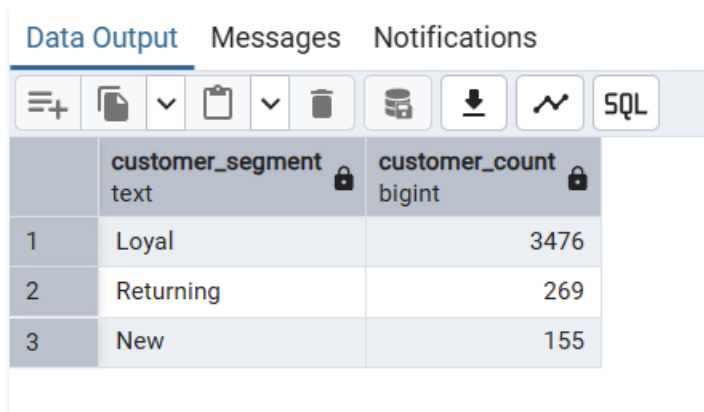
	item_purchased text	discount_percentage numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Total rows: 5 Query complete 00:00:00.100

10.7 Business Question 7 – Customer Segmentation

7. How can customers be segmented into New, Returning and Loyal based on previous purchases?

```
SELECT
CASE
    WHEN previous_purchases <= 2 THEN 'New'
    WHEN previous_purchases <= 5 THEN 'Returning'
    ELSE 'Loyal'
END AS customer_segment,
COUNT(*) AS customer_count
FROM mytable
GROUP BY customer_segment
ORDER BY customer_count DESC;
```



	customer_segment text	customer_count bigint
1	Loyal	3476
2	Returning	269
3	New	155

10.8 Business Question 8 – Top Products by Category

8. What are the top three most purchased products within each category?

```
WITH product_counts AS (
    SELECT category, item_purchased, COUNT(*) AS purchase_count
    FROM mytable
    GROUP BY category, item_purchased
),
ranked_products AS (
    SELECT category, item_purchased, purchase_count,
        RANK() OVER (
            PARTITION BY category
            ORDER BY purchase_count DESC
        ) AS product_rank
    FROM product_counts
)
SELECT category, item_purchased, purchase_count
FROM ranked_products
WHERE product_rank <= 3
ORDER BY category, product_rank;
```

	category text	item_purchased text	purchase_count bigint
1	Accessori...	Jewelry	171
2	Accessori...	Sunglasses	161
3	Accessori...	Belt	161
4	Clothing	Blouse	171
5	Clothing	Pants	171
6	Clothing	Shirt	169
7	Footwear	Sandals	160
8	Footwear	Shoes	150
9	Footwear	Sneakers	145
Total rows: 11		Query complete 00:00:00.070	

10.9 Business Question 9 –

9. Repeat Buyers and Subscription Are repeat buyers with more than five previous purchases more likely to subscribe?

SELECT

CASE WHEN previous_purchases > 5

THEN 'Repeat Buyer'

ELSE 'Non-Repeat Buyer'

END AS customer_type,

COUNT(*) AS total_customers,

SUM(CASE WHEN subscription_status = 'Yes' THEN 1 ELSE 0 END)

AS subscribed_customers,

ROUND(

100.0 * SUM(CASE WHEN subscription_status = 'Yes' THEN 1 ELSE 0 END)

/ COUNT(*), 2

) AS subscription_percentage

FROM mytable

GROUP BY customer_type;

	buyer_type text	total_customers bigint	subscribed_customers bigint	subscription_percentage numeric
1	Repeat Buyer	3476	958	27.56
2	Non-Repeat Buy...	424	95	22.41

10.10 Business Question 10 – Revenue by Age Group

10. What is the revenue contribution of each age group?

```
SELECT age_group,  
       SUM(purchase_amount) AS total_revenue,  
       ROUND(  
         100.0 * SUM(purchase_amount) /  
         SUM(SUM(purchase_amount)) OVER (), 2  
       ) AS revenue_contribution_percentage  
FROM mytable  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Data Output Messages Notifications			
	age_group text	total_revenue numeric	revenue_contribution_percentage numeric
1	56+	65256	28.00
2	46-55	45619	19.57
3	26-35	44342	19.02
4	36-45	43234	18.55
5	18-25	34630	14.86

11. POWER BI VISUALIZATION AND DAX

After SQL analysis, the PostgreSQL database was connected to Power BI. The mytable data was imported into Power BI and used as the source for the dashboard.

11.1 DAX Measures

DAX was used to create reusable measures for key performance indicators.

Total Customers =

`DISTINCTCOUNT(mytable[customer_id])`

Total Revenue =

`SUM(mytable[purchase_amount])`

Average Purchase Amount =

`AVERAGE(mytable[purchase_amount])`

Average Review Rating =

`AVERAGE(mytable[review_rating])`

The exact field names should match the Power BI model.

11.2 Recommended Dashboard Visuals

Visual	Purpose
Total Customers KPI	Shows customer volume.
Total Revenue KPI	Shows total purchase revenue.
Average Purchase Amount KPI	Shows average customer transaction value.
Average Review Rating KPI	Shows average product/customer rating.
Revenue by Gender	Compares revenue contribution by gender.
Revenue by Age Group	Shows demographic revenue contribution.
Customer Segment Chart	Compares New, Returning and Loyal customers.
Top Products Chart	Highlights popular/high-performing products.
Subscription Analysis	Compares subscribers and non-subscribers.
Discount Analysis	Shows discount usage and product discount rates.

12. DASHBOARD DESIGN

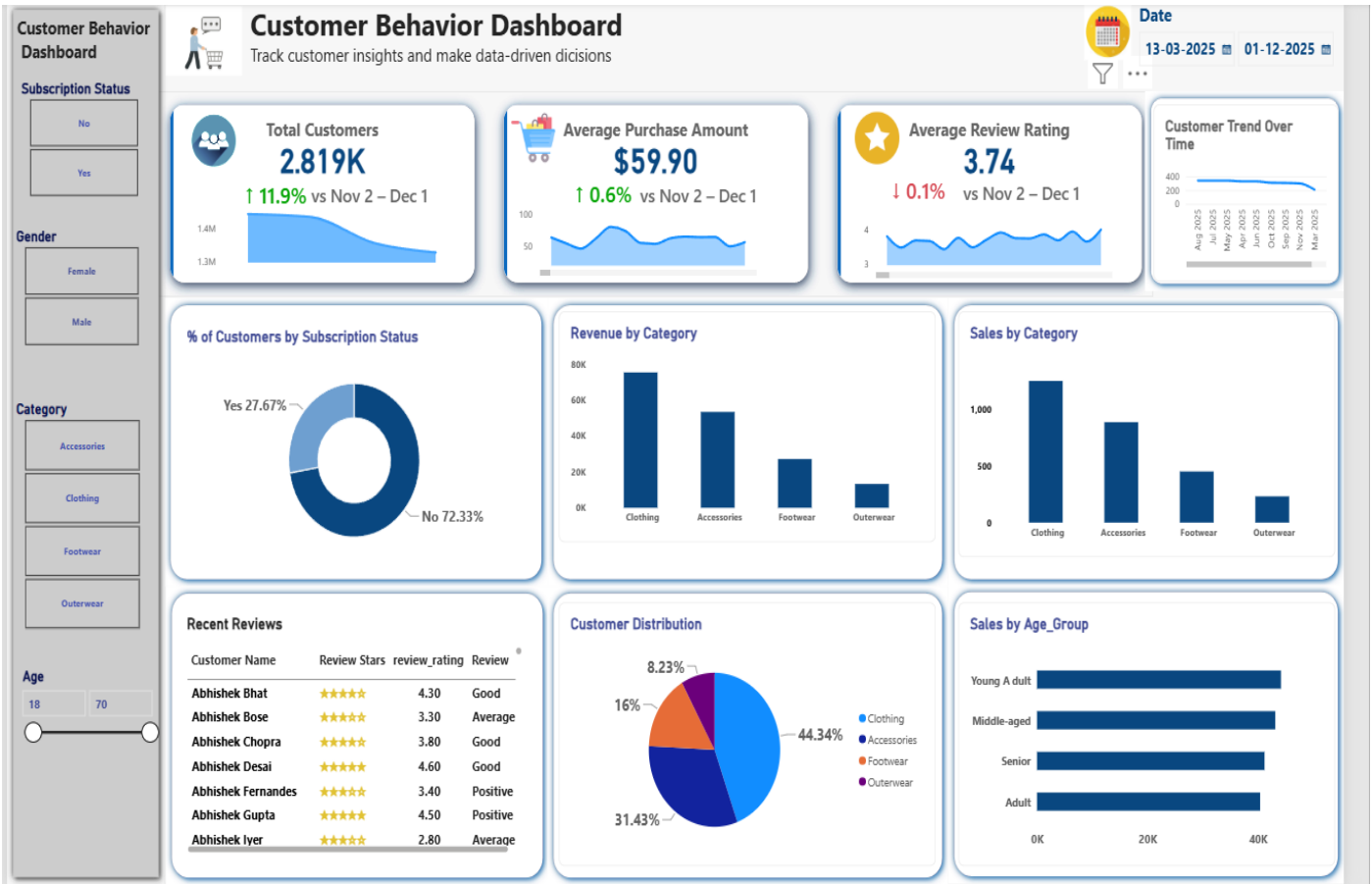
The Power BI dashboard is the presentation layer of the project. It combines KPI cards, charts and filters so stakeholders can explore customer behaviour interactively.

12.1 Dashboard Structure

- Top row: Total Customers, Total Revenue, Average Purchase Amount and Average Review Rating.
- Middle section: Revenue by Gender, Revenue by Age Group and Customer Segmentation.
- Lower section: Product performance, subscription behaviour, discount analysis and purchase patterns.
- Slicers: Category, Gender, Age Group, Subscription Status, Shipping Type and other relevant dimensions.

12.2 Dashboard Objective

The dashboard is designed to convert analytical results into an easy-to-understand management view. Instead of reading raw SQL outputs, stakeholders can interact with charts and filters to identify patterns, compare customer groups and support marketing and product decisions.



13. KEY FINDINGS AND BUSINESS RECOMMENDATIONS

The exact numerical findings should be taken from the SQL query results and Power BI dashboard. The following recommendations are based on the analytical questions and are intended to guide interpretation rather than invent results.

13.1 Customer Engagement

Use customer segments based on previous purchases to differentiate communication. New customers can receive onboarding campaigns, returning customers can receive repeat-purchase incentives, and loyal customers can receive loyalty benefits.

13.2 Discount Strategy

Identify products and customer groups with high discount usage. Discounts should be evaluated not only by usage rate but also by purchase value and revenue contribution so that promotions do not unnecessarily reduce margins.

13.3 Subscription Strategy

Compare the spending and subscription rates of repeat buyers with non-repeat buyers. If repeat customers show stronger subscription behaviour, targeted subscription offers can be used to improve retention.

13.4 Product Strategy

Use top-rated and most-purchased products to support product promotion, inventory planning and category-level merchandising.

13.5 Demographic Marketing

Age-group revenue contribution can be used to identify high-value demographic segments and tailor marketing messages, offers and product campaigns.

13.6 Shipping Strategy

Compare Standard and Express customer spending to understand whether faster shipping is associated with higher-value purchases. This can inform shipping promotions and service design.

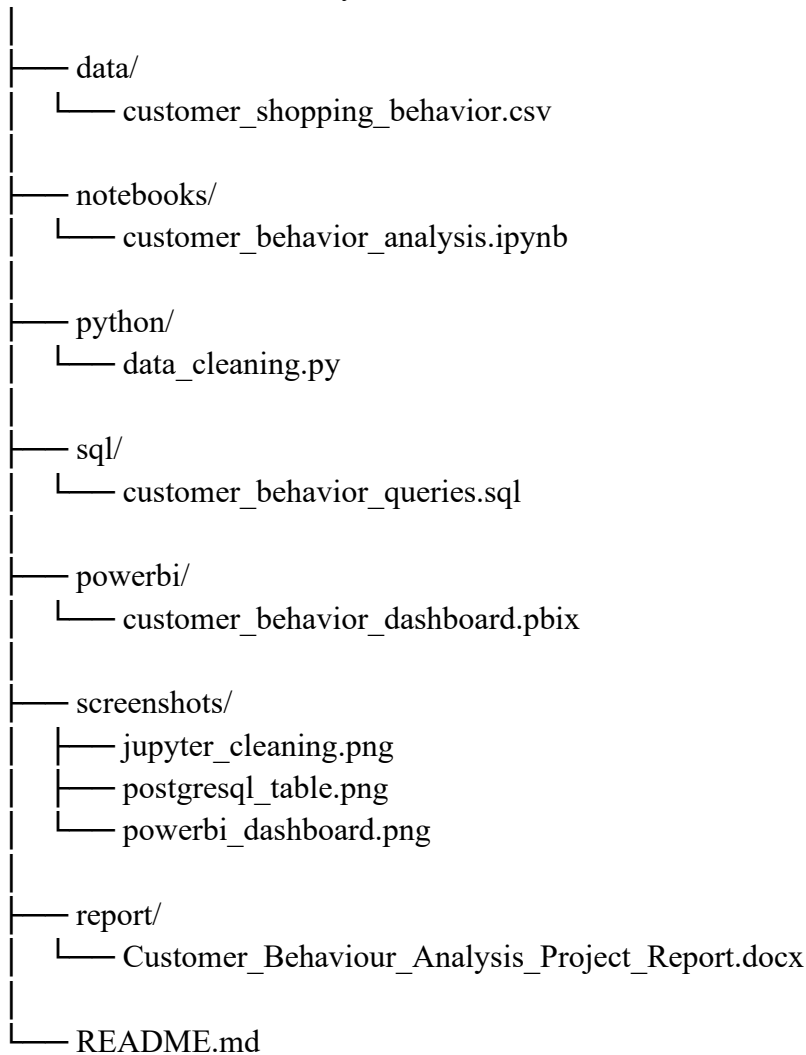
13.7 Dashboard-Driven Decision Making

The Power BI dashboard provides management with a single interactive view of customer behaviour. Regular dashboard review can support faster identification of changing customer patterns.

14. GITHUB REPOSITORY STRUCTURE

The project should be maintained in a clean GitHub repository so that the Python preprocessing, SQL analysis, documentation and dashboard-related assets can be reviewed independently.

Customer-Behaviour-Analysis/



README CONTENT

Project overview and business problem.

Dataset description.

Tools and technologies.

Data-cleaning workflow.

Database setup.

SQL business questions.

Power BI dashboard description.

Instructions for running the project.

Screenshots and key findings.

15. LIMITATIONS AND FUTURE SCOPE

15.1 Limitations

The analysis depends on the fields and quality of the available dataset.

The current project mainly performs descriptive and diagnostic analysis.

The dashboard reflects the data loaded into PostgreSQL and requires refresh/update when new data becomes available.

No predictive machine-learning model is included in the current version.

Business recommendations should be validated against actual company cost, margin and operational data.

15.2 Future Scope

Customer churn prediction using machine learning.

RFM-based customer segmentation.

Customer lifetime value prediction.

Product recommendation systems.

Review sentiment analysis using NLP.

Sales and revenue forecasting.

Automated ETL pipelines.

Scheduled database and Power BI refresh.

Real-time customer analytics.

Advanced marketing campaign response analysis.

16. CONCLUSION

The Customer Behaviour Analysis project successfully implements a complete data analytics pipeline from raw CSV data to an interactive Power BI dashboard. Python and Pandas were used for data inspection, cleaning and feature engineering. PostgreSQL provided structured storage for the cleaned data, while SQL was used to answer ten practical business questions. Power BI and DAX transformed the analytical results into interactive visualizations and KPIs.

The project demonstrates practical skills in data preparation, relational databases, SQL analytics, business problem solving and data visualization. Most importantly, the workflow connects technical analysis with business objectives: understanding customers, improving engagement, optimizing promotions and supporting product and marketing decisions.

17. APPENDIX – IMPORTANT PROJECT CODE

A. Dataset Import

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

B. Data Exploration

```
df.head()
df.info()
df.describe(include='all')
df.isnull().sum()
```

C. Review Rating Imputation

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

D. Column Standardization

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
```

E. Age Group

```
labels = ['Young Adult','Adult','Middle-aged','Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

F. Purchase Frequency

```
frequency_mapping = {'Fortnightly':14,'Weekly':7,'Monthly':30,
'Quarterly':90,'Bi-Weekly':14,'Annually':365,'Every 3 Months':90}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

G. PostgreSQL Load

```
table_name = "mytable"
df.to_sql(table_name, engine, if_exists="replace", index=False)
```

H. Important SQL Concepts Used

GROUP BY and aggregate functions such as SUM, AVG and COUNT.

Subqueries for comparing values with the overall average.

CASE expressions for customer segmentation.

CTEs for multi-step product analysis.

RANK() window function for category-wise product ranking.

Window aggregation for revenue contribution percentage.

